

How do we determine if a D.E is Linear or non-Linear? 2

① D.E is Linear, if there are no products of the function $y(t)$ and its derivatives and neither the function $y(t)$ nor its derivatives occur to any power other than the first power.

② only $y(t)$ and $y'(t)$ are used to determine if the D.E is Linear.

Examples: classify whether the D.E is Linear or Non-Linear. Then determine its degree.

① $(1-x)y'' - 4xy' + 5y = \cos(x)$ (non homogeneous)

This is second order and Linear. or we say the order of the D.E is 2.

② $x y'''(x) - (y'(x))^4 + y = 0$. (homogeneous) → Non-Linear.

This D.E is of order 3 and non-Linear. Note $y'(x)$ is raised to a power greater than 1.

③ $t^5 y^{(4)} - t^3 y'' + 6y = 0$ (homogeneous)

This is Linear D.E of order 4.

④ $y'' + y' + y = \cos(y)$ → Non-Linear.

This is Non Linear D.E of order 2.

Note $\cos(y) = \sum_{n=0}^{\infty} \frac{(-1)^n y^{2n}}{(2n)!} = 1 - \frac{y^2}{2} + \frac{y^4}{4!} - \dots$

So y is raised to power greater than 1.

③

⑤ $y'' = \sqrt{1+(y')^2}$

This is non-linear D.E of order 2.

Note $(y')^2$ (nonlinear).

⑥ $y'' = \sqrt{y}$ is also non linear. why?

$y'' = (y)^{\frac{1}{2}} \rightarrow$ non linear

⑦ $\frac{d^2 y}{dt^2} = \frac{-5}{y^2}$ problem.

This is second order D.E and non linear. why?

$y'' \cdot y^2 = -5$. So y^2 is non linear

Defn: A solution to D.E on an interval (a,b) is any function $y(t)$ which satisfies the D.E in question on the interval (a,b) .

Ex: Show that $y(x) = x^{-\frac{3}{2}}$ is a solution to the D.E

$4x^2 y'' + 12xy' + 3y = 0$ for $x > 0$.

Solution: $y'(x) = -\frac{3}{2} x^{-\frac{5}{2}}$ and $y''(x) = (-\frac{3}{2})(-\frac{5}{2}) x^{-\frac{7}{2}}$

$y''(x) = \frac{15}{4} x^{-\frac{7}{2}}$. Substitute these into the D.E.

$4x^2 \left[\frac{15}{4} x^{-\frac{7}{2}} \right] + 12x \left[-\frac{3}{2} x^{-\frac{5}{2}} \right] + 3x^{-\frac{3}{2}} =$

$= 15x^{-\frac{3}{2}} - 18x^{-\frac{3}{2}} + 3x^{-\frac{3}{2}} = 0. (\checkmark)$

Hence $y(x) = x^{-3/2}$ is a solution.

Integrating Factor Method: We use this method when the D.E is linear first order non-homogeneous:

$$\boxed{\frac{dy}{dt} + P(t)y = g(t)} \dots (1)$$

where $P(t)$ and $g(t)$ are continuous functions.

suppose there is a function say $M(t)$ which we will call "integration factor". we will learn how to find it. Multiply equ(1) by $M(t)$ to get:

$$\boxed{M(t) \frac{dy}{dt} + M(t)P(t)y = M(t)g(t)} \dots (2).$$

Assume that $\boxed{M(t) \cdot P(t) = M'(t)} \dots (3).$

Then equ(2) becomes:

$$\boxed{M(t) \frac{dy}{dt} + M'(t)y = M(t)g(t)} \dots (4)$$

But $M(t)y'(t) + M'(t)y(t) = (M(t) \cdot y(t))'$. Then equ(4) $\Rightarrow$

$$\boxed{(M(t) \cdot y(t))' = M(t)g(t)} \dots (5)$$

Integrate equ(5) with respect to "t".

$$\int (M(t) \cdot y(t))' dt = \int M(t)g(t) dt.$$

$$\therefore M(t) \cdot y(t) = \int M(t)g(t) dt + c$$

$$\therefore y(t) = \frac{1}{M(t)} \left[\int M(t)g(t) dt + c \right] \dots (6)$$

(2)

Now we shall find $\mu(t)$:

From equ(3) $\frac{\mu'(t)}{\mu(t)} = P(t) \Rightarrow [\ln(\mu(t))]' = P(t)$

$$\ln(\mu(t)) = \int P(t) dt + K$$

$$\therefore \boxed{\mu(t) = e^{\left[\int P(t) dt + K \right]}} \Rightarrow \mu(t) = e^K \cdot e^{\left(\int P(t) dt \right)}$$

$$\therefore \boxed{\mu(t) = C \cdot e^{\left[\int P(t) dt \right]}}$$

* Steps to use the integrating factor method: *

① Put the D.E in standard form: $y'(t) + P(t)y(t) = g(t)$.

② Find $\mu(t) = e^{\left[\int P(t) dt \right]}$

③ $y(t) = \frac{1}{\mu(t)} \left[\int \mu(t) \cdot g(t) dt + C \right]$

④ If we have I.C, we use it to Find C.

Ex(1): Find the general solution to $x^3 y' - 3x^2 y = x^6 e^x$. 

Solution: ① $y' - \frac{3x^2}{x^3} y = \frac{x^6 e^x}{x^3} \Rightarrow \boxed{y' - \frac{3}{x} y = x^3 e^x}$
 $\hookrightarrow P(x) = -\frac{3}{x} \hookrightarrow g(x)$

② $\mu(x) = e^{\int -\frac{3}{x} dx} = e^{-3 \ln(x)} = e^{\ln(x^{-3})} = x^{-3} = \frac{1}{x^3}$

③ $y(x) = \frac{1}{\frac{1}{x^3}} \left[\int \frac{1}{x^3} (x^3 e^x) dx + C \right] = x^3 \left[\int e^x dx + C \right]$

$\therefore \boxed{y(x) = x^3 \cdot e^x + Cx^3}$

(Exact Differential Equations)

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Defn: The equation $M(x,y)dx + N(x,y)dy = 0$ is an exact D.E if there exist a function $F(x,y)$ with continuous partial derivatives such that

$\frac{\partial F}{\partial x} = M(x,y)$ and $\frac{\partial F}{\partial y} = N(x,y)$. Then general solution is $F(x,y) = C$

Remark: $\frac{\partial M}{\partial y} = \frac{\partial^2 F}{\partial y \partial x} = \frac{\partial^2 F}{\partial x \partial y} = \frac{\partial N}{\partial x}$

Note $\frac{\partial^2 F}{\partial y \partial x} = \frac{\partial^2 F}{\partial x \partial y}$ only when the partial derivative are continuous.

Fact: If $M(x,y)$ and $N(x,y)$ have continuous partial derivatives. Then $M(x,y)dx + N(x,y)dy = 0$ ^{is exact} if and only if,

How to Find the solution?

$$M_y = N_x$$

$$\frac{\partial F}{\partial x} = M(x,y) \Rightarrow \int \frac{\partial F}{\partial x} dx = \int M(x,y) dx$$

$$\Rightarrow F(x,y) = \int M(x,y) dx + g(y)$$

$$\text{OR } \frac{\partial F}{\partial y} = N(x,y) \Rightarrow \int \frac{\partial F}{\partial y} dy = \int N(x,y) dy$$

$$F(x,y) = \int N(x,y) dy + h(x)$$

Remark: $F_x dx + F_y dy = 0 \Rightarrow F_x + F_y \frac{dy}{dx} = 0 \Rightarrow \frac{d}{dx} [F(x,y(x))] = 0$

Hence $F(x,y) = C$.

Ex: solve the I.V.P: $(2xy - 9x^2) + (2y + x^2 + 1) \frac{dy}{dx} = 0$ (11)

I.C: $y(0) = -3$

Solution: $(2xy - 9x^2)dx + (2y + x^2 + 1)dy = 0$

$M(x,y) = 2xy - 9x^2$ and $N(x,y) = 2y + x^2 + 1$

$M_y = 2x$ and $N_x = 2x \Rightarrow \boxed{M_y = N_x}$. Hence exact D.E

To find $F(x,y)$, we can choose either $\boxed{F_x = M}$ or $\boxed{F_y = N}$

If we choose $F_x = M(x,y) \Rightarrow F(x,y) = \int F_x dx = \int M(x,y) dx$

$F(x,y) = \int (2xy - 9x^2) dx = x^2y - 3x^3 + g(y)$

so $\boxed{F(x,y) = x^2y - 3x^3 + g(y)}$ --- (1) we need to find $g(y)$?

$F_y = x^2 + g'(y) = N(x,y) = 2y + x^2 + 1$

$\Rightarrow g'(y) = 2y + 1 \Rightarrow \boxed{g(y) = y^2 + y + K}$. substitute in eqn(1)

$F(x,y) = x^2y - 3x^3 + y^2 + y + K = C$

Thus $y(x^2 + 1) + y^2 - 3x^3 = C - K = \boxed{C}$

$\boxed{y(x^2 + 1) + y^2 - 3x^3 = C}$ (General solution)

But $y(0) = -3$. That is $x=0, y=-3$

$-3((0)^2 + 1) + (-3)^2 - 3(0)^3 = C \Rightarrow \boxed{C = 6}$

Hence $y(x^2 + 1) + y^2 - 3x^3 = 6$

OR $\boxed{y(x^2 + 1) + y^2 - 3x^3 - 6 = 0}$ (Actual implicit solution)

Note: $\boxed{y(x) = \frac{-(x^2 + 1) \pm \sqrt{x^4 + 12x^3 + 2x^2 + 25}}{2}}$ (Actual explicit solution)

Ex: Solve the I.V.P: $2xy^2 + 4 = 2(3 - x^2y)y'$. I.C: $y(-1) = 8$ 12

Solution: $(2xy^2 + 4)dx + (2x^2y - 6)dy = 0$

$M(x,y) = 2xy^2 + 4$ and $N(x,y) = 2x^2y - 6 \Rightarrow \boxed{M_y = 4xy, N_x = 4xy}$
Hence exact D.E.

Here we choose $F_y = N(x,y) \Rightarrow F(x,y) = \int (2x^2y - 6)dy$

$\boxed{F(x,y) = x^2y^2 - 6y + h(x)} \dots (1)$

$F_x = 2xy^2 + h'(x) = M(x,y) = 2xy^2 + 4$

$\Rightarrow h'(x) = 4 \Rightarrow \boxed{h(x) = 4x + K}$. substitute in equ(1)

$\boxed{x^2 + y^2 - 6y + 4x = C} \dots (2)$

But $y(-1) = 8 \Rightarrow x = -1, y = 8$.

$(-1)^2 + 64 - 6(8) + 4(-1) = C \Rightarrow \boxed{C = 12}$

Thus $x^2 + y^2 - 6y + 4x = 12$

$\boxed{x^2 + y^2 - 6y + 4x - 12 = 0}$

$y(x) = \frac{6 \pm \sqrt{36 - 4x^2(4x - 12)}}{2x^2}$

But $y(-1) = 8$

$\therefore \boxed{y(x) = \frac{3 + \sqrt{9 + 12x^2 - 4x^3}}{x^2}}$